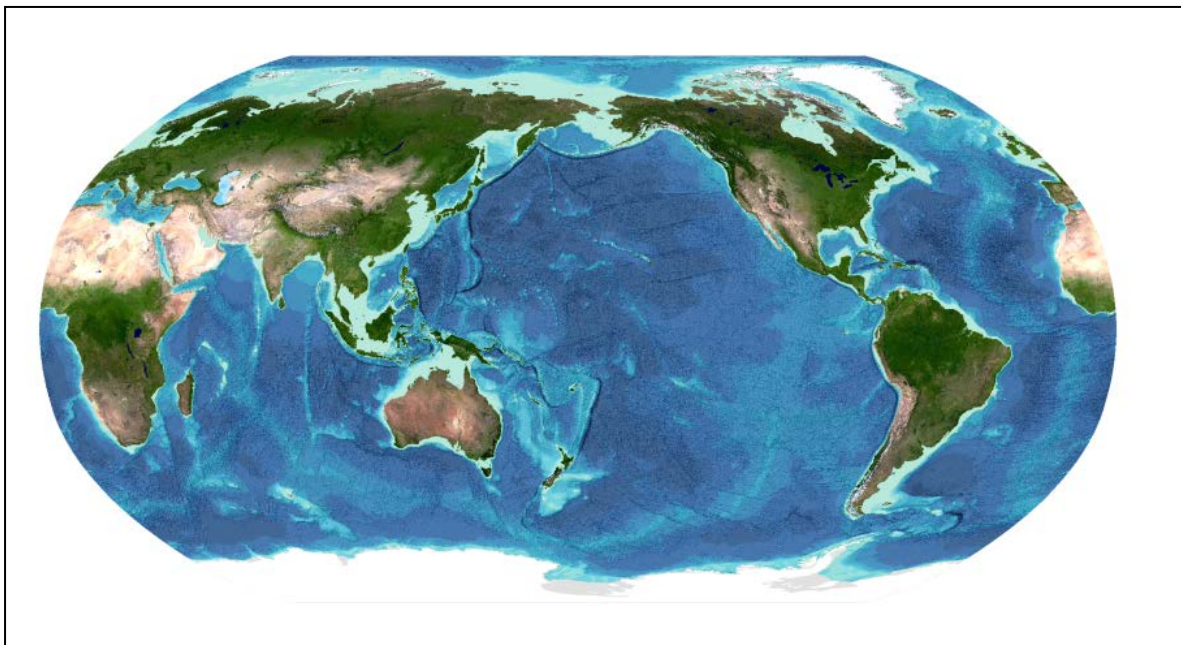


GEBCO's global gridded bathymetric data sets



1 Introduction

This documentation provides information on GEBCO's global bathymetric grids developed since 2019, through The Nippon Foundation-GEBCO [Seabed 2030 Project](#). This is a collaborative project between The [Nippon Foundation](#) of Japan and [GEBCO](#). The Seabed 2030 Project aims to bring together all available bathymetric data to produce the definitive map of the world ocean floor and make it available to all.

The Nippon Foundation is a non-profit philanthropic organisation active around the world. GEBCO is an international group of mapping experts developing a range of bathymetric data sets and data products, operating under the joint auspices of the International Hydrographic Organization ([IHO](#)) and the Intergovernmental Oceanographic Commission ([IOC](#)) of UNESCO.

The GEBCO_2026 Grid, was published in April 2026 and is a global terrain model for ocean and land, providing elevation data, in meters, on a 15 arc-second interval grid.

For polar regions data are available in polar projection co-ordinates.

GEBCO's grids are available to download according to the Terms of Use provided in Section 7 below.

1.1 Seabed 2030 Data Center structure

Since 2019, GEBCO's global grids have been developed, on behalf of GEBCO, by the Seabed 2030 [Centers](#), comprised of four Regional Centers and a Global Center.

The Regional Centers are responsible for championing mapping activities, assembling and compiling bathymetric information and collaborating with existing mapping initiatives in

their regions. The Global Center is responsible for producing and delivering centralized GEBCO products, such as bathymetric grids.

1.2 Seabed 2030 Centers

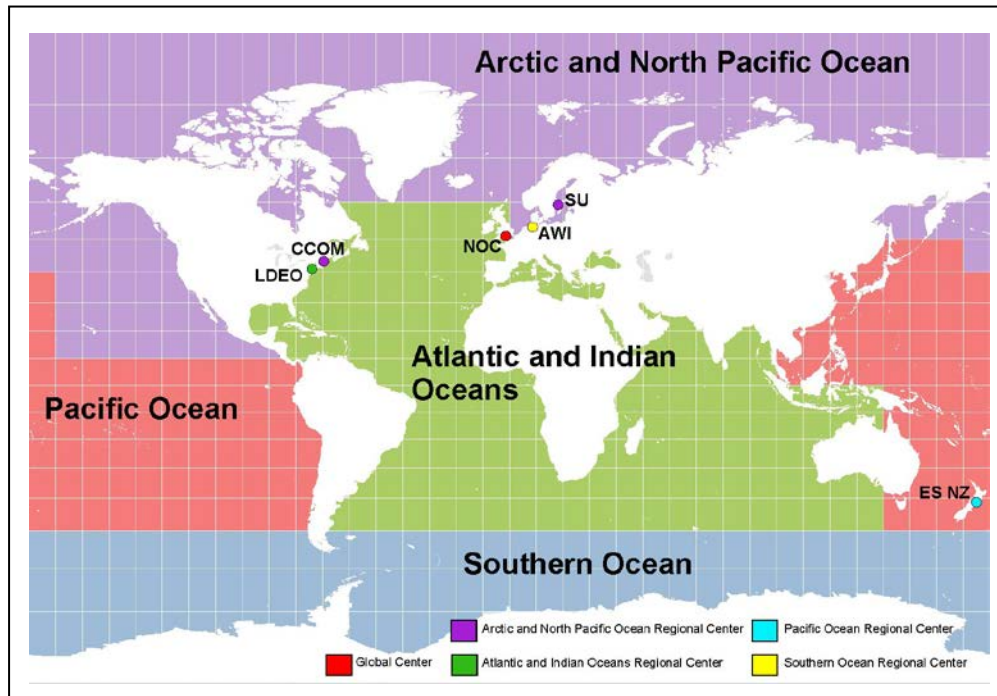


Figure 1. Locations of the Seabed 2030 Centers

- Southern Ocean - hosted at the Alfred Wegener Institute (AWI), Germany
- Pacific Ocean – Earth Sciences, New Zealand (ES NZ)
- Atlantic and Indian Oceans - hosted at the Lamont Doherty Earth Observatory (LDEO), Columbia University, USA
- Arctic and North Pacific Oceans - hosted at Stockholm University (SU), Sweden and the Center for Coastal and Ocean Mapping at the University of New Hampshire (UNH), USA
- Global Center - hosted at the British Oceanographic Data Centre (BODC), National Oceanography Centre (NOC), UK

2 Grid development

GEBCO's global grids are continuous terrain models for ocean and land with a spatial resolution of 15 arc seconds. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

Please note that for since the 2022 release of the GEBCO grid, data are available in polar projection co-ordinates for polar regions.

Below, information is given about the generation of each release of the GEBCO grid.

2.1 GEBCO_2026

The GEBCO_2026 Grid was released in April 2026. It provides global coverage of elevation data, in meters, on a 15 arc-second interval geographic grid and consists of 43200 rows x

86400 columns, giving 3,732,480,000 data points. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

The grid uses as a 'base' Version 2.8 of the SRTM15+ data set [Tozer et al 2019] largely between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography - predictions based on satellite-derived gravity data. This release of the SRTM15+ data set uses a new highly-accurate gravity field data set from the Surface Water and Ocean Topography (SWOT) satellite (Yu et al., 2024) and machine learning methods to produce a bathymetric model (Sandwell et al., 2025).

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2026 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2026 Grid from these regional grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

The sparse regional grids have been included on to the base grid using a 'remove-restore' blending procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and Jakobsson, 2011). This is a two-stage process of computing the difference between the new data and the 'base' grid and then gridding the difference and adding the difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set.

For the polar data sets supplied in the form of complete grids these data sets were included using feather blending techniques from Global Mapper software version 26.2 from Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid releases.

Information on the data sets included in the grid is given in our data contributors list on GEBCO's web site.

Ice-surface elevation and under-ice topography

The GEBCO_2026 grid is made available in two versions, containing:

- land and ice surface elevation information
- under-ice topography information for Greenland and Antarctica

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 6 (Morlighem, M. et al. 2025) and data based on MEaSUREs BedMachine Antarctica, Version 3 (Morlighem, M. 2022).

Land Data

The land data in the GEBCO Grid are taken directly from SRT15+ V2.8 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSURES BedMachine Antarctica, Version 3 (Morlighem, M. 2022).

For areas north of 60°N, land data are largely taken from IceBridge BedMachine Greenland, Version 6 (Morlighem, M. et al. 2025); Svalbard Archipelago: Terrengmodell Svalbard (S0 Terrengmodell) (<https://doi.org/10.21334/npolar.2014.dce53a47>); Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) (<https://doi.org/10.5066/F7J38R2N>); Global Self-consistent, Hierarchical, High-resolution Geography Database (GSHHG) (<https://doi.org/10.1029/96JB00104>).

A list of data sources is given on GEBCO's web site.

Data set reference

GEBCO Compilation Group (2026) GEBCO 2026 Grid (doi: 10.5285/4f68d5c7-45eb-f999-e063-7086abc036fa)

Data in polar projection co-ordinates

To accompany the GEBCO_2026 release of the GEBCO grid, data sets have been made available for polar regions in polar stereographic projection co-ordinates. The data are taken from two of the regional data sets that are included in the global GEBCO_2026 Grid.

These data sets have been developed in polar projection co-ordinates.

Arctic Ocean region (approx. north of 64°N)

The data for this region are taken from version 5.2 of the International Bathymetric Chart of the Arctic Ocean (IBCAO) and included as a regional compilation in the GEBCO_2026 Grid.

The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters. Further information about the IBCAO can be found on GEBCO's web site (<https://www.gebco.net/data-products/gridded-bathymetry-data/arctic-ocean>).

The data set was developed at the Department of Geological Sciences, Stockholm University, Sweden, acting as part of the Arctic and North Pacific Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84 / IBCAO Polar Stereographic (EPSG: 3996)

Southern Ocean region (south of 50°S)

This data set is included as a regional compilation in the GEBCO_2026 Grid. The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters, on a 500m interval grid.

Further information about the IBCSO can be found on GEBCO's web site (<https://www.gebco.net/data-products/gridded-bathymetry-data/southern-ocean>). The data set was developed at the Alfred Wegener Institute (AWI), Germany, acting as the Southern Ocean Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84/ IBCSO Polar Stereographic projection (EPSG: 9354).

Version 2.7 of SRTM15_plus is used in some regions as a 'base grid' with other data sets included on top of this base grid.

2.2 GEBCO_2025

The GEBCO_2025 Grid was released in July 2025. It provides global coverage of elevation data, in meters, on a 15 arc-second interval geographic grid and consists of 43200 rows x 86400 columns, giving 3,732,480,000 data points. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

The grid uses as a 'base' Version 2.7 of the SRTM15+ data set [Tozer et al 2019] largely between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography.

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2025 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2025 Grid from these regional grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

The sparse regional grids have been included on to the base grid using a 'remove-restore' blending procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and Jakobsson, 2011). This is a two-stage process of computing the difference between the new data and the 'base' grid and then gridding the difference and adding the difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set.

For the polar data sets supplied in the form of complete grids these data sets were included using feather blending techniques from Global Mapper software version 23.0.1 from Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid releases.

Information on the data sets included in the grid is given in our data contributors list on GEBCO's web site.

Ice-surface elevation and under-ice topography

The GEBCO_2025 grid is made available in two versions, containing:

- land and ice surface elevation information
- under-ice topography information for Greenland and Antarctica

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 5 (Morlighem, M. et al. 2017) and data based on MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020).

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ V2.7 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020). For areas north of 60°N, land data are largely taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set (Danielson, J.J., and Gesch, D.B., 2011). For the Svalbard region, land data are taken from Terrengmodell Svalbard (S0 Terrengmodell), Norwegian Polar Institute (2014). A list of data sources is given on GEBCO's web site.

Data set reference

GEBCO Compilation Group (2025) GEBCO 2025 Grid (doi:10.5285/37c52e96-24ea-67ce-e063-7086abc05f29)

Data in polar projection co-ordinates

To accompany the GEBCO_2025 release of the GEBCO grid, data sets have been made available for polar regions in polar stereographic projection co-ordinates. The data are taken from two of the regional data sets that are included in the global GEBCO_2025 Grid.

These data sets have been developed in polar projection co-ordinates.

Arctic Ocean region (approx. north of 64°N)

The data for this region are taken from version 5.1 of the International Bathymetric Chart of the Arctic Ocean (IBCAO) and included as a regional compilation in the GEBCO_2025 Grid.

The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters. Further information about the IBCAO can be found on GEBCO's web site (https://www.gebco.net/data_and_products/gridded_bathymetry_data/arctic_ocean/).

The data set was developed at the Department of Geological Sciences, Stockholm University, Sweden, acting as part of the Arctic and North Pacific Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84 / IBCAO Polar Stereographic (EPSG: 3996)

Southern Ocean region (south of 50°S)

This data set is included as a regional compilation in the GEBCO_2025 Grid. The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters, on a 500m interval grid.

Further information about the IBCSO can be found on GEBCO's web site (https://www.gebco.net/data_and_products/gridded_bathymetry_data/southern_ocean/). The data set was developed at the Alfred Wegener Institute (AWI), Germany, acting as the Southern Ocean Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84/ IBCSO Polar Stereographic projection (EPSG: 9354).

Version 2.2.5 of SRTM15_plus is used in some regions as a 'base grid' with other data sets included on top of this base grid.

2.3 GEBCO_2024

The GEBCO_2024 Grid was released in July 2024. It provides global coverage of elevation data, in meters, on a 15 arc-second interval geographic grid and consists of 43200 rows x 86400 columns, giving 3,732,480,000 data points. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

The grid uses as a 'base' Version 2.6 of the SRTM15+ data set [Harper & Sandwell, 2024; Tozer et al 2019] largely between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography.

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2024 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2024 Grid from these regional grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

For the 2020 and 2021 releases of the GEBCO grid, the data sets provided as sparse grids by the Regional Centers were included on to the base grid without any blending. This led to discontinuities at the boundary between the regional grids and the base grids in some areas, largely in regions where the base grid is not constrained by measured data, i.e. areas of large differences between the data sets.

Since 2022, the sparse regional grids have been included on to the base grid using a 'remove-restore' blending procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and Jakobsson, 2011). This is a two-stage process of computing the difference between the new data and the 'base' grid and then gridding the difference and adding the difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set. However, please note that there may be differences between the 2022 and 2021 grid in regions outside areas of measured data due to the grid merging process.

For the polar data sets supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 23.0.1 made available by Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid and notified to the Global Center.

The GEBCO_2024 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. Information on the data sets included in the grid is given in our [data contributors list](#).

Ice-surface elevation and under-ice topography

The GEBCO_2024 grid is made available in two versions, containing:

- land and ice surface elevation information
- under-ice topography information for Greenland and Antarctica

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 5 (Morlighem, M. et al. 2017) and data based on MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020).

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ V2.6 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020). For areas north of 60°N, land data are largely taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set (Danielson, J.J., and Gesch, D.B., 2011). For the Svalbard region, land data are taken from Terrengmodell Svalbard (S0 Terrengmodell), Norwegian Polar Institute (2014). A list of [data sources](#) is given on GEBCO's web site.

Data set reference

GEBCO Compilation Group (2024) GEBCO 2024 Grid (doi:10.5285/1c44ce99-0a0d-5f4f-e063-7086abc0ea0f)

Data in polar projection co-ordinates

To accompany the GEBCO_2024 release of the GEBCO grid, data sets have been made available for polar regions in polar stereographic projection co-ordinates. The data are taken from two of the regional data sets that are included in the global GEBCO_2024 Grid.

These data sets have been developed in polar projection co-ordinates.

Arctic Ocean region (approx. north of 64°N)

The data for this region are taken from version 5.0 of the International Bathymetric Chart of the Arctic Ocean (IBCAO) and included as a regional compilation in the GEBCO_2024 Grid.

The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters. Further information about the IBCAO can be found on GEBCO's web site (https://www.gebco.net/data_and_products/gridded_bathymetry_data/arctic_ocean/).

The data set was developed at the Department of Geological Sciences, Stockholm University, Sweden, acting as part of the Arctic and North Pacific Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84 / IBCAO Polar Stereographic (EPSG: 3996)

Southern Ocean region (south of 50°S)

This data set is included as a regional compilation in the GEBCO_2024 Grid. The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters, on a 500m interval grid.

Further information about the IBCSO can be found on GEBCO's web site (https://www.gebco.net/data_and_products/gridded_bathymetry_data/southern_ocean/). The data set was developed at the Alfred Wegener Institute (AWI), Germany, acting as the Southern Ocean Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84/ IBCSO Polar Stereographic projection (EPSG: 9354).

2.3 GEBCO_2023

The GEBCO_2023 Grid was released in April 2023. It provides global coverage of elevation data, in meters, on a 15 arc-second interval grid and consists of 43200 rows x 86400 columns, giving 3,732,480,000 data points. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells. Please note that the GEBCO_2023 Type Identifier (TID) Grid was updated in January 2024 to correct errors in TID coding in the North Pacific region and central Mediterranean Sea region.

The grid uses as a 'base' Version 2.5.5 of the SRTM15+ data set [Tozer et al., 2019] between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography. It uses predicted depths based on the V32 gravity model [Sandwell et al., 2019].

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2023 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2023 Grid from these regional grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

For the 2020 and 2021 releases of the GEBCO grid, the data sets provided as sparse grids by the Regional Centers were included on to the base grid without any blending. This led to discontinuities at the boundary between the regional grids and the base grids in some areas, largely in regions where the base grid is not constrained by measured data, i.e. areas of large differences between the data sets.

For the 2023 and 2022 Grids, the sparse regional grids have been included on to the base grid using a 'remove-restore' blending procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and Jakobsson, 2011). This is a two-stage process of computing the difference between the new data and the 'base' grid and then gridding the difference and adding the difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set. However, please note that there may be differences between the 2022 and 2021 grid in regions outside areas of measured data due to the grid merging process.

For the polar data sets supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 23.0.1 made available by Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid and notified to the Global Center.

The GEBCO_2023 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. Information on the data sets included in the grid is given in our [data contributors list](#).

Ice-surface elevation and under-ice topography

The GEBCO_2023 grid is made available in two versions, containing:

- land and ice surface elevation information
- under-ice topography information for Greenland and Antarctica

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 4.6 (Morlighem, M. et al. 2017) and data based on MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020).

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ V2.5.5 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020). For areas north of 60°N, land data are largely taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set (Danielson, J.J., and Gesch, D.B., 2011).

2.4 GEBCO_2022

The GEBCO_2022 Grid was released in June 2022. It is a continuous, global terrain model for ocean and land with a spatial resolution of 15 arc seconds. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

The grid uses as a 'base' Version 2.4 of the SRTM15+ data set between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography. It uses predicted depths based on the V31 gravity model [Sandwell et al., 2019]. This version of SRTM15+ is similar to version 2.1 [Tozer et al., 2019] but also includes:

- data from 905 new multibeam sonar cruises, archived at the US National Centers for Environmental Information.
- nine large composite grids from IFREMER for the Gulf of Aden (Hebert et al., 2001), the Lesser Antilles volcanic arc (Talbot & Loubrieu, 2020), French Guiana margin (Loubrieu, 2019a), the Rodrigues triple junction (Mendel et al., 2000), the Kerguelen plateau (Loubrieu, 2019b), Reunion island (Sisavath et al., 2011), Saint-Paul and Amsterdam Islands (Loubrieu et al., 2020), the Southwest Indian ridge (Sauter & Mendel, 2000), and the North Fiji basin (Ruellan, 2001).

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2022 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2022 Grid from these regional data grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

For the 2020 and 2021 releases of the GEBCO grid, the data sets provided as sparse grids by the Regional Centers were included on to the base grid without any blending. This led to discontinuities at the boundary between the regional grids and the base grids in some areas, largely in regions where the base grid is not constrained by measured data, i.e. areas of large differences between the data sets.

For the 2022 Grid, the sparse regional grids have been included on to the base grid using a 'remove-restore' blending procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and Jakobsson, 2011). This is a two-stage process of computing the difference between the new data and the 'base' grid and then gridding the difference and adding the difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set. However, please note that there may be differences between the 2022 and 2021 grid in regions outside areas of measured data due to the grid merging process.

For the polar data sets supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 23.1.0 made available by Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid and notified to the Global Center.

The GEBCO_2022 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. Information on the data sets included in the grid is given in our data [contributors list](#).

Ice-surface elevation and under-ice topography

The GEBCO_2022 grid is made available in two versions, containing:

- land and ice surface elevation information
- under-ice topography information for Greenland and Antarctica

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 4.6 (Morlighem, M. et al. 2017) and data based on MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020).

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ V2.4 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020). For areas north of 60°N, land data are largely taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set (Danielson, J.J., and Gesch, D.B., 2011).

Data in polar projection co-ordinates

To accompany the GEBCO_2022 release of the GEBCO grid, data sets have been made available for polar regions in polar stereographic projection co-ordinates. The data are taken from two of the regional data sets that are included in the global GEBCO_2022 Grid.

These data sets have been developed in polar projection co-ordinates.

Arctic Ocean region (approx. north of 64°N)

The data for this region are taken from version 4.2 of the International Bathymetric Chart of the Arctic Ocean (IBCAO) and included as a regional compilation in the GEBCO_2022 grid.

The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters, on a 200m interval grid. Further information about the IBCAO can be found on GEBCO's web site.

The data set was developed at the Department of Geological Sciences, Stockholm University, Sweden, acting as part of the Arctic and North Pacific Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84 / IBCAO Polar Stereographic (EPSG: 3996)

Attribution: Jakobsson, M., Mayer, L.A., Bringensparr, C. et al. The International Bathymetric Chart of the Arctic Ocean Version 4.0. Scientific Data 7, 176 (2020). <https://doi.org/10.1038/s41597-020-0520-9>. Included as part of the GEBCO_2022 Grid, GEBCO Bathymetric Compilation Group 2022 (2022). The GEBCO_2022 Grid. doi: doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c

Southern Ocean region (south of 50°S)

This data set was developed as part of version 2.0 of the International Bathymetric Chart of the Southern Ocean - IBCSO v2 (Dorschel et al. ,2022) and included as a regional compilation in the GEBCO_2022 grid. The data set can be downloaded in polar projection co-ordinates and represents elevation values, in meters, on a 500m interval grid.

Further information about the IBCSO can be found on GEBCO's web site. The data set was developed at the Alfred Wegener Institute (AWI), Germany, acting as the Southern Ocean Regional Center for the Nippon Foundation-GEBCO Seabed 2030 Project.

Projection details: WGS 84/ IBCSO Polar Stereographic projection (EPSG: 9354)

Attribution: Dorschel, B. et al. The International Bathymetric Chart of the Southern Ocean Version 2. Scientific Data 9, 275, <https://doi.org/10.1038/s41597-022-01366-7> (2022). Included as part of the GEBCO_2022 Grid, GEBCO Bathymetric Compilation Group 2022 (2022). The GEBCO_2022 Grid. doi: doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c

2.5 GEBCO_2021

The GEBCO_2021 Grid was released in July 2021. It uses as a 'base' Version 2.2 of the SRTM15+ data set between latitudes of 50° South and 60° North. This data set is a fusion of land topography with measured and estimated seafloor topography. This version of SRTM15+ is similar to version 2.1 [Tozer et al., 2019] with minor updates. Version 2.2 uses predicted depths based on the V29 gravity model [Sandwell et al., 2019] and approximately 400 small areas containing suspect data were visually identified and removed from the grid.

The SRTM15+ base grid has been augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers to produce the GEBCO_2021 Grid.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2021 Grid from these regional data grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

The data sets provided as sparse grids by the Regional Centers were included on to the base grid without any blending, i.e. grid cells in the base grid were replaced with data from the sparse grids. This was with aim of avoiding creating edge effects, ‘ridges and ripples’, at the boundaries between the sparse grids and base grid during the blending process used previously. In addition, this allows a clear identification of the data source within the grid, with no cells being ‘blended’ values. Routines from the Generic Mapping Tools ([GMT](#)) system were used to do the merging of the data sets.

For the polar data sets, and the adjoining North Sea area, supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 11.0 made available by Blue Marble Geographics.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid and notified to the Global Centre.

The GEBCO_2021 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. Access a list of the [data sets included in the GEBCO_2021 Grid](#).

Ice-surface elevation and under-ice topography

The primary GEBCO_2021 grid contains land and ice surface elevation information - as provided for previous GEBCO grid releases. In addition, for the 2021 release a version of the grid is available with under-ice topography information for Greenland and Antarctica.

The information for ice-surface elevation and under-ice topography/bathymetry is taken from IceBridge BedMachine Greenland, Version 3 (Morlighem, M. et al. 2017) and data based on MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020).

Land Data

The land data in the GEBCO_2021 Grid are taken directly from SRTM15+ V2.2 data set for all areas outside the Polar regions – see the SRTM15_plus data set documentation for more information.

South of 60°S, the land/ ice-surface elevation topography is largely determined from MEaSUREs BedMachine Antarctica, Version 2 (Morlighem, M. et al 2020). For areas north of 60°N, land data are largely taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set (Danielson, J.J., and Gesch, D.B., 2011).

2.6 GEBCO_2020

The GEBCO_2020 Grid was released in May 2020. It uses as a ‘base’ Version 2 of the SRTM15+ data set (Tozer et al, 2019). This data set is a fusion of land topography with measured and estimated seafloor topography. It is augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers.

The Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, for their areas of responsibility. These regional grids were then provided to the Global Center.

For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of ‘sparse grids’, i.e. only grid cells that contain data were populated. For

the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2020 Grid from these regional data grids was carried out at the Global Center, with the aim of producing a seamless global terrain model.

In contrast to the development of the previous GEBCO grid, GEBCO_2019, the data sets provided as sparse grids by the Regional Centers were included on to the base grid without any blending, i.e. grid cells in the base grid were replaced with data from the sparse grids. This was with aim of avoiding creating edge effects, 'ridges and ripples', at the boundaries between the sparse grids and base grid during the blending process used previously. In addition, this allows a clear identification of the data source within the grid, with no cells being 'blended' values. Routines from Generic Mapping Tools (GMT) system were used to do the merging of the data sets.

For the polar data sets, and the adjoining North Sea area, supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 11.0 made available by Blue Marble Geographic.

Some additional edits were made to the final grid to remove erroneous values identified in the previous grid and notified to the Global Centre.

The GEBCO_2020 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. Access a [list of the data sets included in the GEBCO_2020 Grid](#).

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ V2 data set for all areas outside the Polar regions.

South of 60°S, the land topography is largely determined from Bedmap2 (Fretwell et al, 2013). For areas north of 60°N, land data are taken from the Global Multi-resolution Terrain Elevation Data 2010 (GMTED2010) data set.

2.7 GEBCO_2019

The GEBCO_2019 Grid was released in April 2019. It uses as a 'base' Version 1 of the SRTM15+ data set (Olson et al, 2014). This data set is a fusion of land topography with measured and estimated seafloor topography. It is largely based on version 11 of SRTM30+ (Becker et al, 2009; Sandwell et al, 2014), augmented with the gridded bathymetric data sets developed by the four Seabed 2030 Regional Centers. The published data were recalculated on a cell-registered grid for use by GEBCO.

The Seabed 2030 Regional Centers have compiled gridded bathymetric data sets, largely based on multibeam data, on a sub-set of the global grid for their areas of responsibility. These regional grids were then provided to the Global Center. For areas outside of the polar regions (primarily south of 60°N and north of 50°S), these data sets are in the form of 'sparse grids', i.e. only grid cells that contain data were populated. For the polar regions, complete grids were provided due to the complexities of incorporating data held in polar coordinates.

The compilation of the GEBCO_2019 Grid from these regional data grids, with some additional source data sets, was carried out at the Global Center, with the aim of producing a seamless global terrain model.

The majority of the work, for the non-polar regions, was done using the 'remove-restore' procedure (Smith and Sandwell, 1997; Becker, Sandwell and Smith, 2009 and Hell and

Jakobsson, 2011). This is a two stage process of computing the difference between the new data and the 'base' grid, gridding the difference and adding this difference back to the existing 'base' grid. The aim is to achieve a smooth transition between the 'new' and 'base' data sets with the minimum of perturbation of the existing base data set.

For the polar data sets, and the adjoining North Sea are, supplied in the form of complete grids these data sets were included using feather blending techniques from GlobalMapper software version 11.0, made available by Blue Marble Geographic.

The GEBCO_2019 Grid includes data sets from a number of international and national data repositories and regional mapping initiatives. [Access information on the data sets included in the GEBCO_2019 Grid.](#)

Land Data

The land data in the GEBCO Grid are taken directly from SRTM15+ for all areas outside the Polar regions.

South of 60°S, the land topography is determined from Bedmap2 (Fretwell et al, 2013).

2.8 Multi-resolution Grids – TEST DATA SET

As detailed above, the global GEBCO grid is made available as a 15 arc-second interval grid. However, in some regions, it is based on data at higher resolutions. To accommodate users who want access to higher resolution data, where it exists, a multi-resolution grid product is being developed.

At the start of the Seabed 2030 project a set of depth-dependent 'resolution goals' were defined to help determine how much of the ocean floor has been mapped (Mayer et al. (2018)). This is effectively an estimation of data density. These resolutions and depth ranges have been used as the template for the structure of the multi-resolution grids.

The table below shows the depth range and grid resolution for the various multi-resolution grids.

Depth range (meters)	Grid resolution (degrees)	Approximate grid resolution (meters)
0–1,500	0.001	100
0 – 3,000	0.002	200
0 – 11,000	0.004	400

Table 1. Depth range and grid resolution for GEBCO multiple resolution products

Test multi-resolution bathymetric grids are available for an area off Australia in the Indian Ocean (supplied by the Seabed 2030 Atlantic and Indian Ocean Regional Center); the North Pacific region off the Hawaiian Islands (supplied by the Seabed 2030 Arctic and North Pacific Regional Center) and the Southwest Pacific Ocean region off New Zealand (supplied by the Pacific Ocean Regional Center).

Only grid cells that are based on measured bathymetry data are populated, other grid cells are left empty.

These data are based on the data submitted for inclusion in the GEBCO_2024 Grid, published July 2024.

A Type Identifier (TID) Grid is currently not available for the test multi-resolution grids.

Please note that the multi-resolution grids are currently made available as test data sets.

3.0 GEBCO Type Identifier (TID) Grid

GEBCO's bathymetric grids are accompanied by a Type Identifier (TID) grid. This data set identifies the type of source data that the corresponding grid cells in the GEBCO Grid are based on.

The table below details the coding of the GEBCO Type Identifier (TID) grid.

TID	Definition
0	Land
Direct measurements	
10	Single beam - depth value collected by a single beam echo-sounder
11	Multibeam - depth value collected by a multibeam echo-sounder
12	Seismic - depth value collected by seismic methods
13	Isolated sounding - depth value that is not part of a regular survey or trackline
14	ENC sounding - depth value extracted from an Electronic Navigation Chart (ENC)
15	Lidar - depth derived from a bathymetric lidar sensor
16	Depth measured by optical light sensor
17	Combination of direct measurement methods
Indirect measurements	
40	Predicted based on satellite-derived gravity data - depth value is an interpolated value guided by satellite-derived gravity data
41	Interpolated based on a computer algorithm - depth value is an interpolated value based on a computer algorithm (e.g. Generic Mapping Tools)
42	Digital bathymetric contours from charts - depth value taken from a bathymetric contour data set
43	Digital bathymetric contours from ENCs - depth value taken from bathymetric contours from an Electronic Navigation Chart (ENC)
44	Depth value at this location is based on bathymetric depths from multiple sources including measured and derived data and included within a gridded data set where interpolation between sounding points is guided by satellite-derived gravity data
45	Predicted based on helicopter/flight-derived gravity data

46	Depth estimated by calculating the draft of a grounded iceberg using satellite-derived freeboard measurement
47	Depth derived from grounded Argo float data
48	Depth measured by animal-borne data loggers
Unknown	
70	Pre-generated grid - depth value is taken from a pre-generated grid that is based on mixed source data types, e.g. single beam, multibeam, interpolation etc.
71	Unknown source - depth value from an unknown source
72	Steering points - depth value used to constrain the grid in areas of poor data coverage

Table 2. Type Identifier (TID) code definitions

4.0 GEBCO Grid, Vertical and horizontal datum

GEBCO's global gridded data sets provide coverage, spanning 89° 59' 52.5"N, 179° 59' 52.5"W to 89° 59' 52.5"S, 179° 59' 52.5"E on a 15 arc-second geographic latitude and longitude grid.

Each grid consists of 43200 rows x 86400 columns, giving 3,732,480,000 data points. The data values are pixel-centre registered i.e. they refer to elevations, in meters, at the centre of grid cells.

The GEBCO grid can be assumed to be relative to WGS84.

GEBCO's global elevation models are generated by the assimilation of heterogeneous data types, assuming all of them to be referred to Mean Sea Level. However, in some shallow water areas, the grid includes data from sources having a vertical datum other than mean sea level.

5.0 Data Dissemination

GEBCO's gridded data sets are made available in a number of different formats as described in the following sections.

Data for the GEBCO Grid are available in each format as a 'one-click' download option from <https://www.gebco.net/data-products/gridded-bathymetry-data#global>

Data for the previous releases are available in netCDF format from <https://www.gebco.net/data-products/historical-data-sets>.

User-defined subsets from all grids can also be accessed, in various formats, using the [download tool](#). The information below details the formats available for data sets accessed through the download tool.

5.1 CF-compliant NetCDF format

NetCDF (Network Common Data Form) is a self-describing, platform independent data format.

GEBCO's global grids of elevation values are available in NetCDF 4 format and conform to the NetCDF Climate and Forecast (CF) Metadata Convention v1.6 (<http://cfconventions.org/>).

Within the NetCDF files, the data are stored as a two-dimensional array of 2-byte integer values of elevation in metres, with negative values for bathymetric depths and positive values for topographic heights.

GEBCO's TID grids are provided in the same NetCDF format, but data are stored as a two-dimensional array of single byte integers.

5.2 Esri ASCII raster format

This is an ASCII format developed for the export/exchange of Esri ARC/INFO rasters. The format consists of a header that gives the geographic extent and grid interval of the data set, followed by the actual grid cell data values.

5.3 Data GeoTIFF

The GeoTiff format contains geo-referencing (geographic extent and projection) information embedded within a Tiff file.

6.0 Data set attribution

If the data sets are used in a presentation or publication then we ask that you acknowledge the source. This should be of the form:

GEBCO_2026 Grid

GEBCO Compilation Group (2026) GEBCO 2026 Grid (doi: 10.5285/4f68d5c7-45eb-f999-e063-7086abc036fa)

GEBCO_2025 Grid

GEBCO Compilation Group (2025) GEBCO 2025 Grid (doi:10.5285/ 37c52e96-24ea-67ce-e063-7086abc05f29)

GEBCO_2024 Grid

GEBCO Compilation Group (2024) GEBCO 2024 Grid (doi:10.5285/1c44ce99-0a0d-5f4f-e063-7086abc0ea0f)

GEBCO_2023 Grid

GEBCO Compilation Group (2023) GEBCO 2023 Grid (doi:10.5285/f98b053b-0cbc-6c23-e053-6c86abc0af7b)

GEBCO_2022 Grid

GEBCO Compilation Group (2022) GEBCO 2022 Grid (doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c)

GEBCO_2021 Grid

GEBCO Compilation Group (2021) GEBCO 2021 Grid (doi:10.5285/c6612cbe-50b3-0cff-e053-6c86abc09f8f)

GEBCO_2020 Grid

GEBCO Compilation Group (2020) GEBCO 2020 Grid (doi:10.5285/a29c5465-b138-234d-e053-6c86abc040b9)

GEBCO_2019 Grid

GEBCO Compilation Group (2019) GEBCO 2019 Grid (doi:10.5285/836f016a-33be-6ddc-e053-6c86abc0788e)

Polar projection grids accompanying the GEBCO_2022 Grid

North polar grid: Jakobsson, M., Mayer, L.A., Bringensparr, C. et al. The International Bathymetric Chart of the Arctic Ocean Version 4.0. Scientific Data 7, 176 (2020). <https://doi.org/10.1038/s41597-020-0520-9>. Included as part of the GEBCO_2022 Grid, GEBCO Bathymetric Compilation Group 2022 (2022). The GEBCO_2022 Grid. doi: doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c

South polar grid: Dorschel, B. et al. The International Bathymetric Chart of the Southern Ocean Version 2. Scientific Data 9, 275, <https://doi.org/10.1038/s41597-022-01366-7> (2022). Included as part of the GEBCO_2022 Grid, GEBCO Bathymetric Compilation Group 2022 (2022). The GEBCO_2022 Grid. doi: doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c

7.0 Terms of use and disclaimer

7.1 Scope

- These terms of use apply to The GEBCO Grid and other GEBCO-derived information products
- For brevity ‘The GEBCO Grid’ is used throughout and should be interpreted as meaning The GEBCO Grid and other GEBCO-derived information products
- Bathymetric Data refers to measurements made by various instruments of the ocean depth, associated ocean properties and the supporting metadata
- Information products are the result of applying algorithms, mathematical techniques, scientific theory and Intellectual Property to data to create useful, derived values.
- As the GEBCO Grid is created by interpolating, applying algorithms and mathematical techniques to bathymetric data, GEBCO considers the GEBCO Grid to be an information product
- GEBCO does not provide the underlying source bathymetric data when distributing the GEBCO Grid

7.2 Terms of use

The GEBCO Grid is placed in the public domain and may be used free of charge.

Use of the GEBCO Grid indicates that the user accepts the conditions of use and disclaimer information given below.

Users are free to:

- Copy, publish, distribute and transmit The GEBCO Grid
- Adapt The GEBCO Grid
- Commercially exploit The GEBCO Grid, by, for example, combining it with other information, or by including it in their own product or application

Users must:

- Acknowledge the source of The GEBCO Grid. A suitable form of attribution is given in the documentation that accompanies The GEBCO Grid.
- Not use The GEBCO Grid in a way that suggests any official status or that GEBCO, or the IHO or IOC, endorses any particular application of The GEBCO Grid.
- Not mislead others or misrepresent The GEBCO Grid or its source.

7.3 Disclaimer

- The GEBCO Grid should NOT be used for navigation or for any other purpose involving safety at sea.
- The GEBCO Grid is made available 'as is'. While every effort has been made to ensure reliability within the limits of present knowledge, the accuracy and completeness of The GEBCO Grid cannot be guaranteed. No responsibility can be accepted by GEBCO, IHO, IOC, or those involved in its creation or publication for any consequential loss, injury or damage arising from its use or for determining the fitness of The GEBCO Grid for any particular use.
- The GEBCO Grid is based on bathymetric data from many different sources of varying quality and coverage.
- As The GEBCO Grid is an information product created by interpolation of measured data, the resolution of The GEBCO Grid may be significantly different to that of the resolution of the underlying measured data.

8.0 Reporting bugs in the GEBCO Grid

While every effort is made to produce an error free grid, some artefacts may still appear in the data set. Please see our [errata web page](#) for information on known errors in the dataset.

If you find any anomalies in the grid then please report them via email (gdacc@seabed2030.org), giving the problem location, and we will investigate.

9.0 References

Becker, J J, D T Sandwell, W H F Smith, J Braud, B Binder, J Depner, D Fabre, J. Factor, S Ingalls, S-H Kim, R Ladner, K Marks, S Nelson, A Pharaoh, R Trimmer, J Von Rosenberg, G Wallace, P Weatherall (2009). Global Bathymetry and Elevation Data at 30 Arc Seconds Resolution: SRTM30_PLUS, *Marine Geodesy*, 32:4, 355-371.

Dorschel, B. et al. The International Bathymetric Chart of the Southern Ocean Version 2. *Scientific Data* 9, 275, <https://doi.org/10.1038/s41597-022-01366-7> (2022).

Danielson, J.J., and Gesch, D.B., 2011, Global multi-resolution terrain elevation data 2010 (GMTED2010): U.S. Geological Survey Open-File Report 2011–1073, 26 p.

Fretwell, P, H D Pritchard D G Vaughan, J L Bamber, N E Barrand, R Bell, C. Bianchi, R G Bingham, D D Blankenship, G Casassa, G Catania, D Callens, H Conway, A J Cook, H F J Corr, D Damaske, V Damm, F Ferraccioli, R Forsberg, S Fujita, Y Gim, P Gogineni, J A Griggs, R C A Hindmarsh, P Holmlund, J W Holt, R W Jacobel, A Jenkins, W Jokat, T Jordan, E C King, J Kohler, W Krabill, M Riger-Kusk, K A Langley, G Leitchenkov, C Leusche, B P Luyendyk, K Matsuoka, J Mouginot, F O Nitsche, Y Nogi, O A Nost, S V Popov, E Rignot, D M Rippin, A Rivera, J Roberts, N Ros, M J Sieger, A M Smith, D

- Steinhage, M Studinger, B Sun, B K Tinto, B C Welch, D Wilson, D A Young, C Xiangbin and A Zirizzotti (2013). Bedmap2: improved ice bed, surface and thickness datasets for Antarctica, *The Cryosphere*, 7, 375-393, 2013, doi.org/10.5194/tc-7-375-2013.
- Gevorgian, J., Sandwell, D. T., Yu, Y., Kim, S. S., & Wessel, P. (2023). Global distribution and morphology of small seamounts. *Earth and Space Science*, 10(4), e2022EA002331
- Harper, Hugh & Sandwell, David. (2024). Global Predicted Bathymetry Using Neural Networks. *Earth and Space Science*. 11. 10.1029/2023EA003199.
- Hébert, H., Deplus, C., Huchon, P., et al (2001). Lithospheric structure of a nascent spreading ridge inferred from gravity data: The western Gulf of Aden. *Journal of Geophysical Research*, 106(B1126), 345363–345426.
- Hell, B and M Jakobsson (2011), Gridding heterogeneous bathymetric data sets with stacked continuous curvature splines in tension, *Mar. Geophys. Res.*, 32(4), 493-501, doi:10.1007/s11001-011-9141-1.
- Jakobsson, M., Mayer, L.A., Bringensparr, C. et al. The International Bathymetric Chart of the Arctic Ocean Version 4.0. *Scientific Data* 7, 176 (2020). <https://doi.org/10.1038/s41597-020-0520-9>
- Loubrieu, B. (2019a). Bathymetry - Kerguelen Islands (synthesis, 2019) cell size 200 m. Ifremer. <https://doi.org/10.12770/1909f2fb-09ee-49e3-8b5f-52b12e53d292>
- Loubrieu, B. (2019b). Bathymetry of the French Guiana margin and Demerara plateau/data compilation, Ifremer. <https://doi.org/10.12770/1f127956-8350-466e-b795-c8a9eb3c218e>
- Loubrieu, B., Jean-Yves, R., & Marcia, M. (2020). Bathymetry around Saint-Paul and Amsterdam islands/data compilation 2020, Ifremer. <https://doi.org/10.12770/342b2892-b0de-4566-bd77-520fb3bf4eaa>
- Mayer, Larry & Jakobsson, Martin & Allen, Graham & Dorschel, Boris & Falconer, Robin & Ferrini, Vicki & Lamarche, Geoffroy & Snaith, Helen & Weatherall, Pauline. (2018). The Nippon Foundation-GEBCO Seabed 2030 Project: The Quest to See the World's Oceans Completely Mapped by 2030. *Geosciences*. 8. 63. 10.3390/geosciences8020063.
- Morlighem, M., C. Williams, E. Rignot, L. An, J. E. Arndt, J. Bamber, G. Catania, N. Chauché, J. A. Dowdeswell, B. Dorschel, I. Fenty, K. Hogan, I. Howat, A. Hubbard, M. Jakobsson, T. M. Jordan, K. K. Kjeldsen, R. Millan, L. Mayer, J. Mouginot, B. Noël, C. O'Cofaigh, S. J. Palmer, S. Rysgaard, H. Seroussi, M. J. Siegert, P. Slabon, F. Straneo, M. R. van den Broeke, W. Weinrebe, M. Wood, and K. Zinglensen. 2017. BedMachine v3: Complete bed topography and ocean bathymetry mapping of Greenland from multi-beam echo sounding combined with mass conservation, *Geophysical Research Letters*. 44. . <https://doi.org/10.1002/2017GL074954>
- Morlighem, M., E. Rignot, T. Binder, D. D. Blankenship, R. Drews, G. Eagles, O. Eisen, F. Ferraccioli, R. Forsberg, P. Fretwell, V. Goel, J. S. Greenbaum, H. Gudmundsson, J. Guo, V. Helm, C. Hofstede, I. Howat, A. Humbert, W. Jokat, N. B. Karlsson, W. Lee, K. Matsuoka, R. Millan, J. Mouginot, J. Paden, F. Pattyn, J. L. Roberts, S. Rosier, A. Ruppel, H. Seroussi, E. C. Smith, D. Steinhage, B. Sun, M. R. van den Broeke, T. van Ommen, M. van Wessem, and D. A. Young. 2020. Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet, *Nature Geoscience*. 13. 132-137. <https://doi.org/10.1038/s41561-019-0510-8>

Morlighem, M. et al. (2025). IceBridge BedMachine Greenland. (IDBMG4, Version 6). [Data Set]. Boulder, Colorado USA. NASA National Snow and Ice Data Center Distributed Active Archive Center. <https://doi.org/10.5067/6B6B225B8V2D>.

Norwegian Polar Institute (2014). Terrengmodell Svalbard (S0 Terrengmodell) [Data set]. Norwegian Polar Institute. <https://doi.org/10.21334/npolar.2014.dce53a47>

Olson, C. J., Becker, J. J. and Sandwell, D T (2014). A new global bathymetry map at 15 arcsecond resolution for resolving seafloor fabric: SRTM15_PLUS, AGU Fall Meeting Abstracts 2014.

Ruellan, E. North Fiji basin (Southwest Pacific) - 1000m. Ifremer. <https://doi.org/10.12770/f651de98-9478-4d42-9dc9-c0ec7a43ae87>

Sandwell, D T., R D Müller, W H F Smith, E Garcia, R Francis (2014). New global marine gravity model from CryoSat-2 and Jason-1 reveals buried tectonic structure, Science, Vol. 346, no. 6205, pp. 65-67, doi: 10.1126/science.1258213.

Sandwell, D.T., Harper, H., Tozer, B. and Smith, W.H., 2019. Gravity field recovery from geodetic altimeter missions. Advances in Space Research.

David T. Sandwell, Benjamin J. Phrampus, Farshad Salajegheh, et al. Bathymetry Prediction with SWOT Gravity Anomaly using Machine Learning Methods: Paper 1 - Model Development. ESS Open Archive. 17 November 2025.

DOI: <https://doi.org/10.22541/essoar.176339961.11752151/v1>

Sauter, D., & Mendel, V. (2000). The Southwest Indian ridge between 65°E and 68°E-120m. Ifremer. <https://doi.org/10.12770/e54561cf-a7aa4336-a935-7cfec6e1cbbb>

Sisavath, E., Babonneau, N., Saint-Ange, F., Bachèlery, P., Jorry, S. J., Deplus, C., et al. (2011). Morphology and sedimentary architecture of a modern volcanoclastic turbidite system: The Cilaos fan, offshore La Réunion Island. Marine Geology, 288(1–4), 1–17. <https://doi.org/10.1016/j.margeo.2011.06.011>

Smith, W H F and D T Sandwell (1997). Global seafloor topography from satellite altimetry and ship depth soundings, Science, v. 277, p. 1957-1962, 26 Sept.

Talbot, S., & Loubrieu, B. Retrieved from <https://w3.ifremer.fr/archimer/doc/00613/72480/https://sextant.ifremer.fr/eng/Data/Catalogue#/metadata/915d04a1-220f-4784-934c-c269a515e5a9>

Tozer, B, Sandwell, D. T., Smith, W. H. F., Olson, C., Beale, J. R., & Wessel, P. (2019). Global bathymetry and topography at 15 arc sec: SRTM15+. Earth and Space Science. 6. <https://doi.org/10.1029/2019EA000658>.

Yu Y, Sandwell DT, Dibarboure G. Abyssal marine tectonics from the SWOT mission. Science. 2024 Dec 13;386(6727):1251-1256. doi: 10.1126/science.ads4472.